# Von Neumann Regular Hyperrings and Applications to Real Reduced Multirings

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Aims and scope

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A multiring is a ring-like structure where the sum is multivalued and a hyperring is a multiring with a strong distributive property. With every multiring we associate a structural presheaf, and when that presheaf is a sheaf, we say that the multiring is geometric. A characterization of geometric von Neumann hyperrings is presented. And we build a von Neumann regular hull for multirings which is used in applications to algebraic theory of quadratic forms. Namely, we describe the functor Q, introduced by M. Marshall in [J. Pure Appl. Alg., **205**, No. 2, 452-468 (2006)], as a left adjoint functor for the natural inclusion of the category of real reduced multirings (similar to real semigroups) into the category of preordered multirings and explore some of its properties. Next, we employ sheaf-theoretic methods to characterize real reduced hyperrings as certain geometric von Neumann regular real hyperrings and construct the functor V , a geometric von Neumann regular hull for a multiring. Finally, we look at some interesting logical and algebraic interactions between the functors Q and V that are useful for describing hyperrings in the

image of the functor Q and that will allow us to explore the theory of quadratic forms for (formally) real semigroups.



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## Additional information

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